MIL-I-27273A
Amendment 3
12 March 1970
SUPERSEDING
Amendment 2
28 January 1969

MILITARY SPECIFICATION

INVERTER, POWER, STATIC, GENERAL SPECIFICATION FOR

This amendment forms a part of Specification MIL-I-27273A, 3 June 1965, and is mandatory for use by all Departments and Agencies of the Department of Defense.

Page 1, paragraph 2.1: Delete "MIL-P-7936," "MIL-E-26144," "MIL-D-70327," and "MIL-STD-826". Add:

"PPP-B-636	Box, Fiberboard
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-781	Reliability Test: Exponential Distribution
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking and Waterproofing with Appropriate Test Methods".

Page 3, paragraph 3.5.3: Delete and substitute:

"3.5.3 Reliability. The inverter shall have a minimum acceptable mean-time-between failures (MTBF) of 1,175 hours demonstrated to a 90 percent confidence level."

Page 3, paragraph 3.5.4, line 3: Delete "and MIL-E-26144".

Page 3, paragraph 3.5.7, line 2: Delete "0.32-inch" and substitute "0.032 inch".

Page 4, paragraph 3.6.1.1, table I: Change heading of first column from "DC surge voltage" to "Peak DC voltage in service".

Page 4, paragraph 3.6.5, line 6: Delete "solid-static" and substitute "solid-state".

Page 5, paragraph 3.6.10: Delete and substitute:

"3.6.10 Output frequency modulation. Variations of frequency resulting from frequency modulation during any 1-minute period shall be within a band of ±2 cps about a mean frequency. Rates of frequency change caused by frequency modulation shall not exceed 13 cps per second."

FSC 6130

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Page 5, paragraph 3.6.11: Delete the last sentence and substitute: "The total harmonic content shall not exceed 4 percent and no individual harmonic shall exceed 3 percent of the fundamental."

Page 5, paragraph 3.6.12, lines 1 and 2: Delete "shall not exceed the limits of MIL-STD-704" and substitute "shall remain within the limits of 54 and 173 volts rms".

Page 5, paragraph 3.6.12.1: Delete and substitute:

"3.6.12.1 Overvoltage spikes. Spikes of microsecond duration on the output voltage shall not exceed a peak of 250 volts. The inverter shall withstand imput voltage spikes in accordance with 4.6.12, table II. Moreover, during design, careful consideration shall be given to the protection of the inverter against any overvoltage spikes that may be internally generated."

Page 6, paragraph 3.6.16, line 2: Delete "not exceed the limits specified in MIL-STD-704" and substitute "be 120 ±1.5 degrees."

Page 6, paragraph 3.6.18(d): Delete "landing".

Page 6, paragraph 3.6.21: Delete and substitute:

3.6.21 Source impedance. The inverter shall be capable of operating satisfactorily when connected to a 28V dc aircraft generator having a current rating of not more than twice the imput current rating of the inverter or 28 amps, whichever is greater."

Page 6, paragraph 3.7, line 3: Delete "MIL-D-70327" and substitute "MIL-STD-100".

Page 6, paragraph 3.8, line 5: Delete "MIL-STD-826" and substitute "MIL-I-6181".

Page 8 and 9, paragraph 4.3.3: Under "(a) Sample no. 1,"item (15) and under "(b) Sample no. 2, items (15) and (9) respectively, delete paragraph reference "4.6.3" and substitute "4.6.18.1."

Page 9, paragraph 4.6.3.1: Delete paragraph number and substitute "4.3.3.1".

Page 10, paragraph 4.6.3: Under "(a)," add:

"(3) At maximum specified input voltage, the load shall be varied from no load to full load in steps of one-fourth of rating. The load shall be suddenly removed and applied for each observation. Voltage and frequency readings shall be taken to accurately plot the voltage and frequency at the minimum lagging and leading power factor loads as specified on the applicable MS."

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Page 12, paragraph 4.6.9.2: Delete and substitute:

"4.6.9.2 Output frequency modulation. The output frequency modulation shall be measured under conditions of no load, half load, and full load and shall not exceed ±2 cps from a mean frequency for a 1-minute period. The rate of frequency change shall not exceed 13 cps per second. Measurements of frequency modulation shall be made by a method approved by the procuring activity."

Page 12, paragraph 4.6.10.1, fifth sentence: Delete and substitute: "The total harmonic content shall not exceed 4 percent and no individual harmonic shall exceed 3 percent of the fundamental."

Page 13, paragraph 4.6.11, line 3: Delete "shall not exceed limits 2 and 3 of MIL-STD-704" and substitute "shall remain within the limits of 54 and 173 volts rms."

Page 13, paragraph 4.6.14.1: Change the title to read "Source impedance." Add the following sentence to the end of the paragraph: "No battery shall be used in the power supply for the inverter."

Page 14, paragraph 4.6.16, line 4: Delete "be within the limits of 120 ±1.5 degrees as specified in MIL-STD-704" and substitute "be 120 ±1.5 degrees."

Page 16, paragraph 4.6.21, lines 3 and 4: Delete "MIL-STD-826" and substitute "MIL-I-6181".

Page 16, add:

"4.6.23 Reliability. The inverter shall pass the qualification test without incurring any relevant failures (see MIL-STD-781, paragraph titled 'Relevant Failures') to demonstrate a minimum acceptable MTBF of 1,175 hours to a 90 percent confidence level."

Page 16, paragraphs 5.1 and 5.2: Delete and substitute:

"5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

"5.1.1 Level A. Inverters shall be preserved in accordance with MIL-P-116, method II, and packaged one each in unit containers conforming to PPP-B-636.

"5.1.2 Level C. Each inverter shall be packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity for immediate use. The level may conform to the supplier's commercial practice if it meets the requirements of this level.

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"5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

"5.2.1 Level A. Inverters preserved and packaged as specified in 5.1 shall be packed in exterior-type shipping containers conforming to the applicable type and class of PPP-B-636. Insofar as practicable, containers shall contain identical quantities and be of minimum cube and tare consistent with the protection required. Container closure and strapping shall be in accordance with the appendix to PPP-B-636.

"5.2.2 Level B. Level B packing shall be the same as level A except the shipping containers shall be of the domestic type.

"5.2.3 Level C. Each inverter shall be packed in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations and may be the supplier's commercial practice if it meets the requirements of this level.

"5.3 Physical Protection. Inverters packed as specified in 5.2.1 and 5.2.2 shall be anchored, clocked, braced, and cushioned in accordance with MIL-STD-1186.

"5.4 Marking. In addition to any special marking required by the contract or order, interior containers and exterior shipping containers shall be marked in accordance with MIL-STD-129."

Page 16, paragraph 6.2, c: Delete and substitute;

"c. Levels of packaging and packing required (see section 5)".

Page 16, paragraph 6.3: Delete and substitute:

"6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products

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covered by this specification. The activity responsible for the Qualified Products List is the Aeronautical Systems Division, Attn: ASNJP-10, Wright-Patterson Air Force Base, Ohio 45433, and information pertaining to qualification of products may be obtained from that activity."

Custodians:
Army - AV
Air Force - 11

Review activities:
Army - AV, EL
Air Force - 82, 85

Preparing activity:
Air Force - 11

Project No. 6130-0087

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